

PROJECT PROBLEM STATEMENT

Bank Transactions & Account Holders

An End-to-End MySQL Data Analytics Project for Beginners

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Domain: Banking & Financial Services

Tool Used: MySQL Workbench 8.0

Level: Beginner / Fresher Portfolio Project

1. Business Context

A mid-sized retail bank maintains savings, current and salary accounts for individual and corporate customers across multiple branches in India. Every day, thousands of transactions flow through channels such as UPI, ATM, POS, NEFT, IMPS, net banking and branch counters. Currently, this data sits in raw transactional form and is not being used to answer basic operational and business questions.

As a Data Analyst joining the bank's analytics team, you have been given two extracts from the core banking system – an account holders file and a transactions file – and asked to set up a proper relational database and produce first-cut insights for the operations and business teams.

2. Problem Statement

The bank's leadership wants to understand three things: (1) who their customers are and how the customer base is distributed across segments, account types and branches; (2) how money is moving through the bank – which channels, branches and categories drive the highest volume and value; and (3) where the operational risks are – failed transactions, pending KYC customers who are still active, and unusually large transactions that may need review.

This information is currently not available in any single place. The goal of this project is to design a clean two-table relational database in MySQL, load the raw CSV extracts into it, and write SQL queries that directly answer these questions.

3. Objectives

- Design a normalized MySQL schema (account_holders and bank_transactions) with an appropriate primary key / foreign key relationship.
- Import the two provided CSV datasets into MySQL Workbench without data loss or mismatched rows.
- Explore the data to confirm row counts, ranges and data quality before analysis.
- Write beginner-friendly SQL queries (joins, GROUP BY, HAVING, CASE WHEN, subqueries, a view) that answer real banking business questions.
- Summarize the findings in a short, decision-ready insight report.

4. Dataset Description

4.1 account_holders.csv (~1,500 rows)

Column	Description
account_id	Unique identifier for each bank account (Primary Key)
customer_name, gender, date_of_birth	Basic customer demographic details
city, state, branch_name, ifsc_code	Location and servicing branch of the account
account_type, customer_segment	Savings / Current / Salary and Retail / Premium / Corporate
account_open_date, initial_deposit, kyc_status	Onboarding details and KYC compliance status

4.2 bank_transactions.csv (~12,000 rows)

Column	Description
transaction_id	Unique identifier for each transaction (Primary Key)
account_id	Links back to account_holders (Foreign Key)
transaction_date, transaction_type	When the transaction happened and whether it was a Credit or Debit
transaction_channel, merchant_category	UPI / ATM / POS / NEFT / IMPS / etc., and spend category for card-based debits
amount, balance_after_transaction	Transaction value and resulting account balance
transaction_status, branch_name	Success / Failed / Pending, and processing branch

5. Scope of Analysis

- Customer base overview – account type, segment and KYC distribution
- Transaction volume and value – credit vs debit, monthly trend, channel and merchant analysis
- Customer-level insights – top customers, branch performance, average balances
- Risk and operational quality – failed transaction rates, high-value transactions, KYC-pending but active customers

6. Tools & Deliverables

Tool used: MySQL Workbench 8.0 (Community Edition).

Deliverables

- Two CSV datasets: account_holders.csv and bank_transactions.csv
- 01_database_and_tables.sql – schema creation script
- 02_data_import_guide.sql – step-by-step CSV import instructions
- 03_exploratory_data_analysis.sql – data validation queries
- 04_business_insights_queries.sql – 18 business-question queries
- Insight Report (Word document) summarizing key findings and recommendations